

Disproof of conjecture 00000009779:

a C^k kernel on $[0, 1]^d$ whose eigenvalues satisfy $\lambda_n \geq \frac{1}{2}n^{-k-5/4}$, so they do not decay like $n^{-k-1-d/2}$

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: Regularity of integral kernels versus eigenvalue decay: power laws for eigenvalues of Sobolev-class kernels. Conjecture: C^k regular kernels satisfy $|\lambda_n| \leq Cn^{-k-1-d/2}$ with optimal C the spherical average of derivative norms of the kernel; for radial kernels the constant is exactly computable and given by the first maximum of the Fourier transform. (kernel regularity-decay optimal constant)

The Chinese text states the same claim with the same exponent $n^{-k-1-d/2}$.

Reading. The conjecture is a conjunction. Its first clause is

- (A) for every $d \geq 1$, $k \geq 0$ and every C^k kernel K on a d -dimensional domain Ω :
 $\exists C \forall n \geq 1 : |\lambda_n| \leq C n^{-k-1-d/2}$,

where $\lambda_1, \lambda_2, \dots$ are the eigenvalues of the integral operator $(T_K \varphi)(x) = \int_{\Omega} K(x, y) \varphi(y) dy$ on $L^2(\Omega)$, listed with multiplicity in order of nonincreasing $|\lambda_n|$. The other two clauses (the optimal C is a spherical average of derivative norms; for radial kernels C is given by the first maximum of the Fourier transform) describe the constant in (A), so they presuppose that such a C exists. We refute (A) for *every* dimension $d \geq 1$ and *every* $k \geq 0$, with $\Omega = [0, 1]^d$ and Lebesgue measure. This refutes the conjunction. The counterexample kernels are also real, symmetric and positive semidefinite. For $d = 1$ the kernel is radial ($K(x, y) = f(|x - y|)$), so the radial clause fails as well: no constant exists to be computed.

Conventions. k ranges over $\mathbb{N} = \{0, 1, 2, \dots\}$. “ C^k kernel” is taken in the strongest sense: K is C^k jointly in (x, y) on all of $\mathbb{R}^d \times \mathbb{R}^d$. Functions in L^2 are complex valued. $\int_0^1$ is the interval integral, which equals the integral over $[0, 1]$. The Lean file writes the dimension as $D = d + 1$ with $d \in \mathbb{N}$, so that $D \geq 1$ and the coordinate x_0 exists.

2 Definitions (as formalized)

All names are in namespace C9779.

- `cube D = [0, 1]^D ⊂ ℝD` (as `Set.univ.pi fun _ => Icc 0 1`).
- `intOp K phi x = ∫[0, 1]D K(x, y) φ(y) dy` (the integral operator T_K).
- `IsOrthonormalEigenfamily K phi mu`: the functions $\varphi_0, \dots, \varphi_{n-1}$ are continuous, satisfy $\int_{[0, 1]^D} \overline{\varphi_i} \varphi_j = \delta_{ij}$, and satisfy $T_K \varphi_i(x) = \mu_i \varphi_i(x)$ at every $x \in [0, 1]^D$.

- **EigenBound K k C**: for every $n \geq 1$ and every such family of n functions there is an i with $|\mu_i| \leq C n^{-(k+1+D/2)}$.
- **PosSemidefKernel K**: $\sum_{a,b} c_a c_b K(x_a, x_b) \geq 0$ for all finite families of points x_a and real coefficients c_a .
- **coef s j** = $(j+1)^{-s}$, **fser s t** = $f_s(t) = \sum_{m \geq 1} m^{-s} \cos(2\pi m t)$, **Kd d s x y** = $f_s(x_0 - y_0)$, **phi d j x** = $e^{2\pi i(j+1)x_0}$.

Lemma 1 (from the ordered eigenvalues to **EigenBound**). *Let T be a linear operator on a complex inner product space, and let $\lambda_1, \lambda_2, \dots$ be its eigenvalues, listed with multiplicity in order of nonincreasing absolute value. Let $c \geq 0$. If $v_1, \dots, v_n$ are orthonormal and $Tv_i = \mu_i v_i$ with $|\mu_i| > c$ for every i , then $|\lambda_n| > c$. Consequently, if $|\lambda_n| \leq C n^{-(k+1+D/2)}$ for all $n \geq 1$, then **EigenBound K k C** holds.*

Proof. Let $\nu_1, \dots, \nu_r$ be the distinct values among the μ_i . Each $\nu_j \neq 0$. The v_i with $\mu_i = \nu_j$ are orthonormal, hence linearly independent, vectors of $\ker(T - \nu_j)$. So the multiplicity of ν_j is at least $\#\{i : \mu_i = \nu_j\}$. Summing over j , at least n entries of the list have absolute value $> c$. Since the list is ordered by nonincreasing absolute value, the first n entries have absolute value $> c$, and in particular $|\lambda_n| > c$.

For the consequence, put $B = C n^{-(k+1+D/2)}$ and take a family as in **EigenBound**. Continuous functions on the compact cube lie in L^2 , and the integral and pointwise identities say that they are orthonormal eigenvectors of T_K on $L^2([0, 1]^D)$. Since $|\lambda_n| \leq B$, we have $B \geq 0$. If every $|\mu_i| > B$, the first part gives $|\lambda_n| > B$, a contradiction. $\square$

So refuting **EigenBound K k C** for every C refutes (A) for K . Lean proves the negation of **EigenBound**. Lemma 1 is the elementary step from that negation to the ordered eigenvalues, and it is not formalized.

3 Proof

Fix $k \geq 0$ and $D \geq 1$, and put $s = k + \frac{5}{4}$. Then $s > k + 1$ and $k + 1 + \frac{D}{2} - s = \frac{D}{2} - \frac{1}{4} \geq \frac{1}{4}$.

Lemma 2 (smoothness; **contDiff_fser**, **contDiff_Kd**). *If $s > k + 1$ then $f_s \in C^k(\mathbb{R})$, and $K(x, y) = f_s(x_0 - y_0)$ is C^k on $\mathbb{R}^D \times \mathbb{R}^D$.*

Proof. The i -th derivative of $t \mapsto (j+1)^{-s} \cos(2\pi(j+1)t)$ is bounded by $(2\pi)^i (j+1)^{i-s}$, which is summable in j for every $i \leq k$, because $s - i > 1$. Termwise differentiation (Mathlib **contDiff_tsum**) gives $f_s \in C^k$. Compose with the linear map $(x, y) \mapsto x_0 - y_0$. $\square$

Lemma 3 (eigen-equation; **eigen_1d**, **eigen_d**). *For $s > 1$, every $m \geq 1$ and every $x \in \mathbb{R}$: $\int_0^1 f_s(x - y) e^{2\pi i m y} dy = \frac{1}{2} m^{-s} e^{2\pi i m x}$. Hence $T_K \varphi_m = \frac{1}{2} m^{-s} \varphi_m$ on $[0, 1]^D$ for $\varphi_m(x) = e^{2\pi i m x_0}$.*

Proof. Write $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$. The j -th term of the integrand is

$$\frac{(j+1)^{-s}}{2} \left(e^{2\pi i(j+1)x} e^{2\pi i(m-j-1)y} + e^{-2\pi i(j+1)x} e^{2\pi i(m+j+1)y} \right).$$

Since $\int_0^1 e^{2\pi i p y} dy = \delta_{p0}$ for $p \in \mathbb{Z}$ (Mathlib **integral_exp_mul_complex**), the j -th integral equals $\frac{1}{2} m^{-s} e^{2\pi i m x}$ if $j+1 = m$, and 0 otherwise. The terms are bounded by $(j+1)^{-s}$, which is summable, so dominated convergence (**intervalIntegral.hasSum_integral_of_dominated_convergence**)

lets us integrate termwise. On the cube, $\int_{[0,1]^D} g(y_0) dy = \int_0^1 g$ for continuous g (`integral_cube_eval`): the cube measure is the product of the measures on $[0, 1]$, which are probability measures, so evaluation at coordinate 0 is measure preserving. $\square$

Lemma 4 (orthonormality; `integral_conj_efun_mul`, `orthonormal_d`). $\int_{[0,1]^D} \overline{\varphi_a} \varphi_b = \int_0^1 e^{2\pi i(b-a)t} dt = \delta_{ab}$.

Lemma 5 (symmetry and positivity; `Kd_symm`, `posSemidef_Kd`). $K(x, y) = K(y, x)$, and for real c_a and points x_a ,

$$\sum_{a,b} c_a c_b K(x_a, x_b) = \sum_{j \geq 0} (j+1)^{-s} \left[\left(\sum_a c_a \cos \omega_j x_{a,0} \right)^2 + \left(\sum_a c_a \sin \omega_j x_{a,0} \right)^2 \right] \geq 0,$$

where $\omega_j = 2\pi(j+1)$.

Proof. f_s is even. Exchange the finite sums with the convergent series and use $\cos(u-v) = \cos u \cos v + \sin u \sin v$. $\square$

Lemma 6 (growth; `growth`). For every $C \in \mathbb{R}$ there is $n \geq 1$ with $C n^{-(k+1+D/2)} < \frac{1}{2}(i+1)^{-s}$ for all $i < n$.

Proof. Choose $n \geq 1$ with $n^{1/4} > 2C$. Put $u = n^{D/2-1/4} \geq n^{1/4} > 2C$. Then $n^{-(k+1+D/2)} = n^{-s}/u$, and $C n^{-s}/u < \frac{1}{2} n^{-s} \leq \frac{1}{2}(i+1)^{-s}$ for $i+1 \leq n$. $\square$

Theorem 7 (`not_eigenBound`, `main`, `main_radial`). For every $D \geq 1$ and $k \geq 0$, the kernel $K(x, y) = f_{k+5/4}(x_0 - y_0)$ on $[0, 1]^D$ is real, symmetric, positive semidefinite and C^k on $\mathbb{R}^D \times \mathbb{R}^D$, and **EigenBound** K k C fails for every C . For $D = 1$, $K(x, y) = f_{k+5/4}(|x - y|)$ is radial.

Proof. Given C , take n from the growth lemma and the family $\varphi_1, \dots, \varphi_n$ with eigenvalues $\mu_m = \frac{1}{2} m^{-s}$. It is an orthonormal eigenfamily, and every $|\mu_m| > C n^{-(k+1+D/2)}$, so **EigenBound** fails at n . For $D = 1$, $f_s(x_0 - y_0) = f_s(|x_0 - y_0|)$ because f_s is even. $\square$

By Lemma 1, $|\lambda_n| \leq C n^{-k-1-D/2}$ fails for every C ; in fact $|\lambda_n| \geq \frac{1}{2} n^{-k-5/4}$ for all $n \geq 1$.

4 Lean formalization and axiom audit

File `lean/Conjecture9779/Basic.lean` (384 lines, import `Mathlib` only). Main statement:

```
theorem main (d k : N) :
  exists K : (Fin (d + 1) -> R) -> (Fin (d + 1) -> R) -> R,
    ContDiff R k (Function.uncurry K) /\ (forall x y, K x y = K y x) /\
    PosSemidefKernel K /\ not (exists C : R, EigenBound K k C)
```

(with $\mathbb{N}$, $\mathbb{R}$ written as `N`, `R`). `main_radial k` gives the same for $D = 1$ with the kernel written as $f(|x_0 - y_0|)$ and $f \in C^k(\mathbb{R})$. Supporting theorems: `contDiff_fser`, `contDiff_Kd`, `eigen_1d`, `eigen_d`, `integral_exp_int`, `integral_conj_efun_mul`, `orthonormal_d`, `integral_cube_eval`, `posSemidef_Kd`, `Kd_symm`, `growth`, `not_eigenBound`.

Axiom audit. `#print` axioms for `C9779.main` and `C9779.main_radial` reports [`propext`, `Classical.choice`, `Quot.sound`]. There is no `sorry`, no `native_decide`, and no added axiom.